THE STACKCALC LIBRARY • CURRICULUM SERIES — BOOK 2 OF 4

HANDS-ON MATH LAB

Building Mathematical Understanding with Physical Blocks, Visual Models & Inside-Out Stack Verification

A self-contained K–12 hands-on student exploration workbook connecting physical manipulative models—Place Value Studio (PLACE-01), Fraction Studio (FRAC-01), and Prime Factor Studio (PRIME-01)—with real-world mathematical concepts, paper-and-pencil predictions, and step-by-step verification on the four-line StackCalc SC-6 and SC-32 calculators.

StackCalc

stackcalc.io

First Edition

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Illustrated Companion Trade Books (History of Mathematics & Stack Thinking)

- Volume I — From Blocks to the Stack: How Hands-On Blocks Make Numbers Click (Grades K–6 / StackCalc SC-6)
- Volume II — Thinking in the Stack: The Art of Calculating Without Parentheses (Grades 7–12, AP & Engineers / StackCalc SC-32)

K–12 Concept-First Curriculum & Student Workbooks (US Letter 8.5" × 11" Editions)

- Book 1 — Teaching with the Stack: Unified K–12 Master Teacher's Guide, Concept Lessons & Complete Answer Keys
- Book 2 — Hands-On Math Lab: K–12 Student Manipulative & Four-Line Stack Exploration Workbook
- Book 3 — Middle School Math Lab: Grades 6–8 Two-Method Student Practice Workbook (900 Problems) & 15-Page Coach Guide
- Book 4 — Stack Speed & Fluency: 15 Progressive Timed Sprints for Inside-Out Mathematical Fluency

Three Published Physical Manipulative Block Studios & Four-Line Stack Calculators (stackcalc.io)

- Place Value Studio (PLACE-01): 6×2 Whole & Decimal Trays, Digit Blocks (0–9) & Stackable Gold [+1] Carry Tokens
- Fraction Studio (FRAC-01): 84.0 mm Proportional Fraction Towers (1 Whole to 1/12) & 12-Socket Comparator Lid
- Prime Factor Studio (PRIME-01): Logarithmic Prime Blocks (2, 3, 5, 7 where equal products stand flush) & GCF/LCM Comparator
- StackCalc Calculators: SC-6 Elementary/Middle School Calculator & SC-32 Scientific Calculator (Hardware, iOS, iPadOS, watchOS & Web)

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HANDS-ON MATH LAB: Building Mathematical Understanding with Physical Blocks, Visual Models & Inside-Out Stack Verification

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Designed to supplement state-mandated mathematics standards—including Common Core State Standards (CCSS-M), Texas Essential Knowledge and Skills (TEKS), Florida B.E.S.T., Virginia SOL, and AP Mathematics / Computer Science—alongside district basal textbooks (Eureka Math / EngageNY, Illustrative Mathematics / Open Up, enVision Mathematics, Big Ideas Math, HMH Into Math, McGraw-Hill Reveal, Bridges in Mathematics, and Singapore Dimensions Math).

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APPENDICES A & B: STUDENT SELF-CHECK ANSWER KEY (PAGE 25) & MASTERY CERTIFICATE (PAGE 26)

Complete all 12 units and check your work on the SC-6 or SC-32 calculator to earn your Stack Mastery Certificate!

Preface: Welcome to the Hands-On Math Lab!

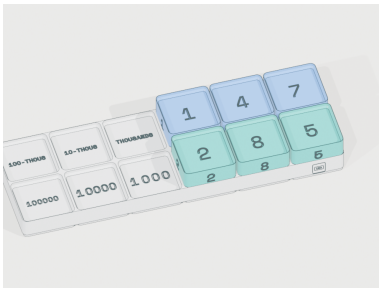
Why your calculator has an [ENTER] key instead of [=]—and how blocks, storybooks, and the stack make math make sense.

1. WHY DOES YOUR CALCULATOR HAVE AN [ENTER] KEY INSTEAD OF AN [=] SIGN?

Have you ever typed a long problem into a regular 1-line calculator, pressed [=], and gotten a strange answer—with no idea where things went wrong? That happens because a 1-line calculator hides your intermediate steps.
Now think about how you add 8 + 5 on paper: first you write 8 on the top line (your Waiting Shelf, Line Y). Next you write 5 below it (your active Workbench, Line X). Only after both numbers are in place do you add (+) to get 13! That step-by-step workspace is called a Stack (or RPN, used by scientists and NASA engineers). On your 4-line calculator (T, Z, Y, X), you press [8] [ENTER] to place 8 on Waiting Shelf Y, type [5] on Workbench X, and press [+] to combine them. Every step stays visible right on the screen!

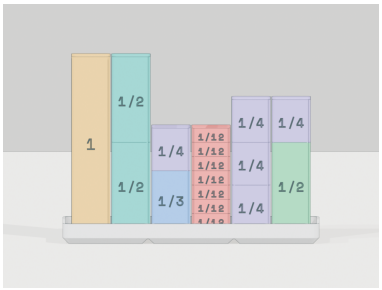
2. YOUR THREE HANDS-ON BLOCK KITS (USED WHEN APPLICABLE)

1. Place Value Blocks (PLACE-01)



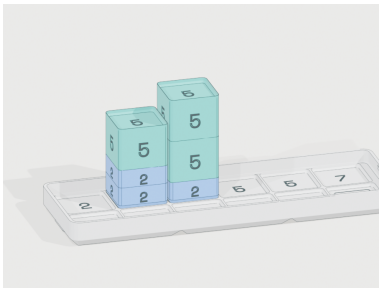
2-Row Tray (Y & X), Digits 0–9 & [+1] Carry

2. Fraction Towers (FRAC-01)



84 mm Proportional Towers (1 to 1/12)

3. Prime Factor Blocks (PRIME-01)



Logarithmic Prime Blocks (2, 3, 5, 7)

3. SOLVE BY HAND FIRST, THEN CHECK ON YOUR CALCULATOR (2-MINUTE WARM-UP)

• Math Comes First: In this workbook, you are learning mathematical concepts—never memorizing calculator buttons! Every unit is 100% self-contained: you build or sketch the math model first, write your step-by-step mathematical prediction on paper, and then use your calculator to verify your work.
• Inside-Out Stack Check: Floor 1 (Line X) is your active Workbench; Floor 2 (Line Y) is your Waiting Shelf; Floors 3 & 4 (Z & T) hold earlier steps so you can check multi-step expressions without parentheses!

- **Try Addition on Your Stack (8 + 5 = 13):** [8] [ENTER] [5] [+]
- **Try a Two-Dot Fraction (1/2 + 1/4 = 3/4):** [•] [1] [•] [2] [ENTER] [•] [1] [•] [4] [+]

The 4-Step Lab Loop in Every Unit: 1. Mathematical Concept → 2. Blocks / Model → 3. Paper Prediction → 4. Check!

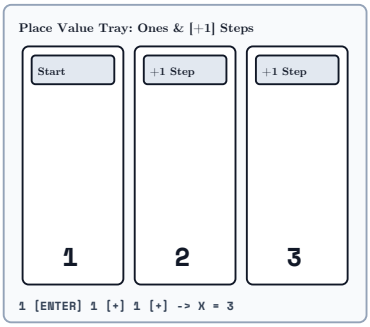
Always solve and write your prediction on paper before pressing calculator keys! After finishing the 4 lab investigations on the right-hand page of any unit, check your answers in Appendix A (Page 25).

Counting, Cardinality & The One-More Step Counter

Hands-On Model: Place Value Studio (Digit Blocks 0–9 & Gold [+1] Step Tokens) • Standards: CCSS.MATH.K.CC.A.1, CCSS.MATH.K.CC.B.4

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (1-to-1 Correspondence: Stepping Up by One):

Counting by ones is repeated addition of the unit 1 (n → n + 1). On the Place Value Tray and the stack, your current count waits first while you bring in 1 more step, then both combine into the new total.



2. HANDS-ON BLOCKS & CONCEPT RULES: HOW COUNTING & PLACE VALUE WORK TOGETHER

- Numbers show quantity (cardinality): the Digit Block in the Ones socket shows how many items you have counted.
- Adding 1 more item steps the count up by 1 (n → n + 1): 1 becomes 2, 2 becomes 3, and 3 becomes 4.
- An empty tray has zero items, represented by the [0] Digit Block in the Ones socket.
- When checking with a partner or teacher on the SC-6, Line Y holds your starting count and Line X adds the next [+1] step.

3. PARTNER / TEACHER STACK CHECK: VERIFYING OUR BLOCKS ON THE SC-6

Hands-On Setup: Place the [1] Digit Block into the Ones socket of the Place Value Tray. Keep Gold [+1] step tokens ready beside the tray. Question: When we place 1 Gold [+1] step token and step the Ones block up by 1, what digit block replaces [1]? (Prediction: [2])

Verification Keystrokes: [1] [ENTER] [1] [+] [1] [+]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[1]	1	First count placed: 1 enters register X
[ENTER]	1	1 copied to register Y: holds the current count
[1]	1	Gold [+1] step token added: 1 enters register X
[+]	2	Press [+]: 1 + 1 = 2! Swap Ones block to [2]
[1]	1	Next Gold [+1] step token added: 1 enters X
[+]	3	Press [+]: 2 + 1 = 3! Tray [3] and screen match perfectly

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Place the [4] Digit Block in the Ones column of your Place Value Tray and add 1 Gold [+1] step token. What total quantity do you have now?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): Row 1 holds [6] and Row 0 holds [2] on your Place Value Tray. Which row has more, and how many more does it have?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 01 Lab: Counting, Cardinality & The One-More Step Counter

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with your Place Value blocks & draw in Box 1. (2) Write the two quantities & math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check with your partner or teacher on the SC-6!

Investigation 1: Place [3] in the top row and [2] in the bottom row of your Place Value Tray. Combine both groups—what is the total?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: You have 5 ones on the tray and take away 2. What Digit Block shows how many ones remain?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: You have 3 blocks on the tray and take away all 3. How many blocks are left, and which Digit Block shows an empty tray?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Start with [1] in the Ones socket and add one more [+1] token three times in a row (1 + 1 + 1 + 1). What count do you reach?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

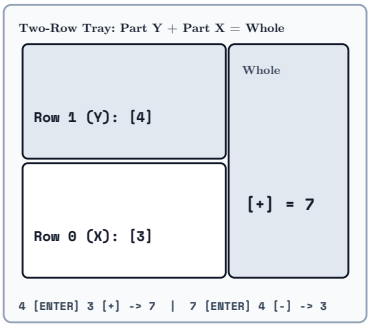
EXIT REFLECTION: When you add one more [+1] token to a full tray of 9, how do the Tens and Ones sockets change?

Part-Part-Whole & Number Bonds on the Place Value Tray

Hands-On Model: Place Value Studio (Two-Row Tray & Digit Blocks 0–9) • Standards: CCSS.MATH.K.OA.A.3, CCSS.MATH.1.OA.C.6

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Composing Numbers: Part + Part = Whole):

Every addition problem composes two independent parts into a single whole (Part 1 + Part 2 = Whole). Placing Part 1 on the top row (Shelf Y) and Part 2 on the bottom row (Shelf X) makes both parts visible before they are joined.



2. HANDS-ON BLOCKS & CONCEPT RULES: PART-PART-WHOLE NUMBER BONDS ON THE TWO-ROW TRAY

- Row 1 (top row) of the Place Value Tray holds Part 1, and Row 0 (bottom row) holds Part 2.
- Putting both parts together (Part 1 + Part 2) composes the Whole.
- To find a missing part when you know the Whole: start with the Whole and subtract the known Part (Whole - Part 1 = Part 2)!

3. PARTNER / TEACHER STACK CHECK: VERIFYING OUR BLOCKS ON THE SC-6

Hands-On Setup: Place the [4] Digit Block in Row 1 (Part 1) and the [3] Digit Block in Row 0 (Part 2) of the Ones column on your Place Value Tray. Question: Combine both parts in the Ones column: 4 + 3 = 7 (swap to the [7] Digit Block).

Verification Keystrokes: `4` `ENTER` `3` `+`

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
`4`	**4**	Part 1 (4) enters register X
`ENTER`	**4**	4 copied up to register Y; represents Row 1 (Part 1)
`3`	**3**	Part 2 (3) enters register X; represents Row 0 (Part 2)
`+`	**7**	Plus combines Part 1 (Y) + Part 2 (X) = Whole (7) in X!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Place [5] in Row 1 (Part 1) and [4] in Row 0 (Part 2) of your Place Value Tray. Combine the two parts to find the Whole.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): A number bond has a Whole of 8 and a known Part of 5 (5 + ? = 8). Find the missing Part 2.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 02 Lab: Part-Part-Whole & Number Bonds on the Place Value Tray

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with your Place Value blocks & draw in Box 1. (2) Write the two quantities & math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check with your partner or teacher on the SC-6!

Investigation 1: To make a Whole of 6, Part 1 is 2 (2 + ? = 6). What is the missing Part 2?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: In a number bond, Part 1 is 7 and Part 2 is 2. What is the Whole when you combine both parts?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: Three parts join together: first combine 2 and 3, then add 4 to that sum. What is the Whole?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Ten birds sit on a fence (Whole = 10). Seven fly to a tree (Part 1 = 7). How many stay on the fence (Part 2)?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

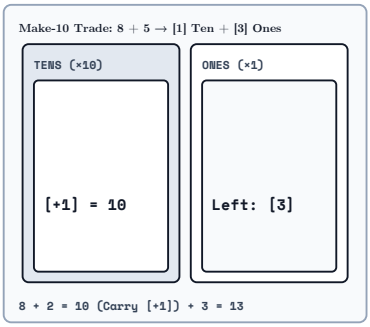
EXIT REFLECTION: How do adding Part 1 + Part 2 and subtracting Whole − Part 1 use the exact same number bond?

Complements to 10 & Make-10 Regrouping with Carry Tokens

Hands-On Model: Place Value Studio (Whole-Number Tray, Digit Blocks 0–9 & Gold [+1] Carry Tokens) • Standards: CCSS.MATH.1.OA.B.4, CCSS.MATH.1.OA.C.6

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Complements to 10: Making Ten & Carry Trade):

Our base-10 system regroups every collection of 10 ones into 1 ten. Finding a number's complement to 10 (10 - 7 = 3) places the target base (10) on Shelf Y and the current ones digit (7) on Workbench X.



2. HANDS-ON BLOCKS & CONCEPT RULES: COMPLEMENTS TO 10 & GOLD [+1] CARRY RULES

- The Ones column holds single Digit Blocks [0] through [9]. The moment a column reaches 10, it regroups!
- 10-for-1 Carry Trade: When the Ones column reaches 10, place a Gold [+1] Carry Token into the Tens column and reset the Ones socket to [0].
- Complements to 10: To find how much more is needed to trigger a Gold [+1] carry, enter 10 [ENTER], then the digit, then [-].
- Bridging Through 10: Decompose the second addend to complete 10 first (e.g., 8 + 5 = 8 + 2 + 3 = 13).

3. PARTNER / TEACHER STACK CHECK: VERIFYING OUR BLOCKS ON THE SC-6

Hands-On Setup: Place the [7] Digit Block into the Ones socket of the Place Value Tray and hold a Gold [+1] Carry Token beside the Tens socket. Question: How many more ones must be added to [7] to reach 10 and trigger the Gold [+1] Carry Token? (7 + 3 = 10).

Verification Keystrokes: [1] [0] [ENTER] [7] [-]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[1]	1	Enter 1
[0]	10	Target ten (10) in X
[ENTER]	10	10 lifted to Y
[7]	7	Current Ones digit (7) in X
[-]	3	10 - 7 = 3: Adding 3 completes 10 and triggers the [+1] Carry!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): The Ones socket holds [6]. How many more ones are needed to complete a full 10 and trade for 1 Ten?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): To add 9 + 6 using the Make-Ten strategy, split 6 into 1 and 5 so you make 10 first (9 + 1 + 5). What is the sum?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 03 Lab: Complements to 10 & Make-10 Regrouping with Carry Tokens

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with your Place Value blocks & draw in Box 1. (2) Write the two quantities & math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check with your partner or teacher on the SC-6!

Investigation 1: You have 4 ones on your tray. How many more ones do you need to make a full 10 (4 + ? = 10)?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: Solve 7 + 5 by bridging through 10: first add 7 + 3 to make 10, then add the remaining 2.

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: The Ones socket holds [9]. How many more ones are needed to complete 10 and regroup into 1 Ten?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Solve 8 + 6 by splitting 6 into 2 and 4 so you make 10 first (8 + 2 + 4).

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

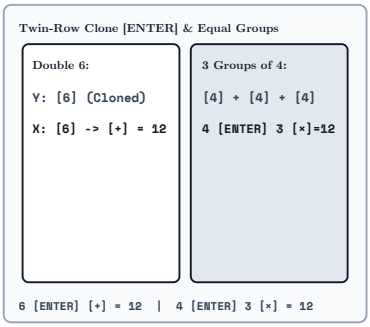
EXIT REFLECTION: Why does bridging through 10 (like 8 + 5 = 8 + 2 + 3) make adding across a ten easier?

Doubles, Near-Doubles & Equal Groups on the Place Value Tray

Hands-On Model: Place Value Studio (Two-Row Tray & Digit Blocks 0–9) • Standards: CCSS.MATH.1.OA.C.6, CCSS.MATH.2.OA.B.2

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Twin-Row Doubles: Adding Two Equal Groups):

Doubling a quantity means adding two identical copies of the same number (n + n = 2n). Seeing matching quantities in the top and bottom rows of the tray anchors mental doubles facts.



2. HANDS-ON BLOCKS & CONCEPT RULES: TWIN-ROW DOUBLES & NEAR-DOUBLES STRATEGIES

• Placing matching Digit Blocks in Row 1 (top) and Row 0 (bottom) of the Place Value Tray models a physical Double (n + n).

• Near-Doubles: When two addends differ by 1 (such as 7 + 8), double the smaller number first (7 + 7 = 14) and add 1 more (14 + 1 = 15).

• On the SC-6, pressing [ENTER] duplicates the number on Line X onto Line Y so pressing [+] checks any double in 3 keystrokes.

3. PARTNER / TEACHER STACK CHECK: VERIFYING OUR BLOCKS ON THE SC-6

Hands-On Setup: Place a [6] Digit Block in Row 1 (top) and a matching [6] Digit Block in Row 0 (bottom) of the Ones column on the Place Value Tray. Question: Observe the matching rows: 6 + 6 = 12 (1 Ten and 2 Ones on the tray).

Verification Keystrokes: [6] [ENTER] [+]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[6]	6	Top row quantity (6) enters register X
[ENTER]	6	ENTER duplicates 6 into register Y! Both rows now on stack
[+]	12	Plus combines top and bottom rows: 6 + 6 = 12!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Place [7] in both rows of your Place Value Tray to build a twin-row double. What is the sum of two 7s (7 + 7)?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): Use the known double 6 + 6 to solve the near-double 6 + 7 (double 6, then add 1).

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 04 Lab: Doubles, Near-Doubles & Equal Groups on the Place Value Tray

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with your Place Value blocks & draw in Box 1. (2) Write the two quantities & math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check with your partner or teacher on the SC-6!

Investigation 1: Two matching rows on the Place Value Tray each hold 8. What is double 8 (8 + 8)?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: Use the double 8 + 8 to solve the near-double 8 + 9 (double 8, then add 1).

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: A carton has two equal rows of 9 eggs. What is double 9 (9 + 9)?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Use the double 5 + 5 to solve the near-double 5 + 6 (double 5, then add 1).

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

EXIT REFLECTION: How can you use the twin-row double 7 + 7 = 14 to find 7 + 8 mentally in one step?

UNIT 05 GUIDED CONCEPT LAB • GRADES 3–4 (SC-6)

The 4-Register Stack: Four Operations & Automatic Lift

Hands-On Model: Place Value Blocks & 4-Register Stack • Standards: CCSS.MATH.3.OA.A.1, CCSS.MATH.3.OA.A.2

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Multi-Step Word Problems Without Parentheses):

In a multi-step expression like (3 × 4) + (2 × 5), each product is a self-contained quantity that must be evaluated before the two groups can be added. On the 4-level stack, the first product (12) waits safely on Shelf Y while the second product (10) is formed on X, requiring zero parentheses.



Place Value Studio (p. vi) • 10-for-1 Carry Trade

2. HANDS-ON BLOCKS & CONCEPT RULES: 4-REGISTER STACK ANATOMY

- Four registers: T (top), Z (third), Y (second/operand), X (screen/bottom).
- ENTER duplicates X into Y and pushes older values up into Z and T.
- Two-number operators (+, -, ×, ÷) combine Y and X, dropping Z to Y and T to Z.
- Automatic Lift: Entering a new number after an operation automatically pushes the result to Y!

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Hands-On Setup: Use Place Value unit blocks to represent quantities alongside the 4-line stack display. Question: Problem: (8 - 3) × 2. Predict: 8 - 3 = 5, then 5 × 2 = 10.

Verification Keystrokes: [8] [ENTER] [3] [-] [2] [×]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[8]	8	8 in X
[ENTER]	8	8 copied to Y
[3]	3	3 entered into X
[-]	5	8 - 3 = 5 in X
[2]	2	Automatic lift! 5 lifts to Y; 2 enters X
[×]	10	5 × 2 = 10 in X!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): A baker bakes 15 cookies, sells 6 of them, and packs the remaining cookies equally into 3 bags. How many cookies go in each bag?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): Evaluate (24 ÷ 6) + 15 by solving the division group first and then adding 15 to that result.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 05 Lab: The 4-Register Stack: Four Operations & Automatic Lift

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your step-by-step paper math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

Investigation 1: Solve (14 - 5) × 3 by finding the difference inside the parentheses first, then multiplying by 3.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: Solve 30 ÷ (2 + 3) by adding the divisor group (2 + 3) first, then dividing 30 by that sum.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: Evaluate (9 + 7) ÷ (8 - 4) by finding the numerator sum and denominator difference first, then dividing.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: A farmer collects 20 eggs, sets aside 2 cracked eggs, and divides the rest equally among 6 cartons. How many per carton?

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

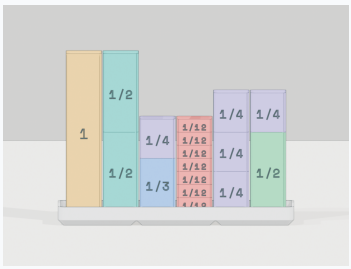
EXIT REFLECTION: In (15 − 6) ÷ 3, why don't we need parentheses when we compute the inner step (15 − 6) first?

Fraction Studio: Height Equivalence & Two-Dot Notation

Hands-On Model: 84 mm Fraction Towers & Twin Comparator Well • Standards: CCSS.MATH.4.NF.A.1, CCSS.MATH.4.NF.B.3

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Mixed Numbers & Height Equivalence: 3/4 + 2 1/2):

Fractions can only be added or subtracted when they share a common unit (denominator), and mixed numbers combine whole units with fractional parts. On the stack, two-dot notation ([.] n [.] d) treats each fraction or mixed number as a single unified value on Y and X.



Fraction Studio (p. vii) • 84 mm Flush Towers

2. HANDS-ON BLOCKS & CONCEPT RULES: TWO-DOT FRACTION ENTRY & TOWER EQUIVALENCE

- One Whole tower stands 84.0 mm tall, dividing cleanly into halves (42 mm), thirds (28 mm), fourths (21 mm), sixths (14 mm), eighths (10.5 mm), and twelfths (7 mm).
- Twin Comparator Well: Stack equivalent fraction towers side-by-side against the 84.0 mm Whole tower to verify equivalence by sight and touch.
- Two-dot entry format: [WHOLE] [.] [NUMERATOR] [.] [DENOMINATOR] (or [a ^{b}/c] on SC-6).
- Proper fractions have 0 whole: press [.] [3] [.] [4] for 3/4.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Hands-On Setup: Place three 1/4 blocks (3 × 21 = 63 mm). Beside them, stack two Whole blocks and one 1/2 block (84 + 84 + 42 = 210 mm). Question: Combine heights: 63 mm + 210 mm = 273 mm. In wholes: 3 1/4 (273 / 84 = 3.25).

Verification Keystrokes: [•] [3] [•] [4] [ENTER] [2] [•] [1] [•] [2] [+]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[•]	0.	Leading dot indicates 0 wholes
[3]	0.3	Numerator = 3
[•]	0.3.	Second dot sets fraction entry mode
[4]	3/4	Denominator = 4: displays 3/4 (or 0.75)
[ENTER]	3/4	3/4 lifted to register Y
[2]	2	Whole = 2
[•]	2.	Dot
[1]	2.1	Numerator = 1
[•]	2.1.	Dot
[2]	2 1/2	Denominator = 2: displays 2 1/2
[+]	3 1/4	Plus combines: 3 1/4 (decimal 3.25)!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Stack a 1/2 tower (42 mm) and a 1/4 tower (21 mm) together. How many 1/4 blocks match their combined height, and what is 1/2 + 1/4?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): Find 2/3 of 3/8 by multiplying (2/3) × (3/8) and simplifying to lowest terms. Which unit fraction tower matches the product?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 06 Lab: Fraction Studio: Height Equivalence & Two-Dot Notation

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your step-by-step paper math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

Investigation 1: Add 1/3 + 1/6 by trading 1/3 for two 1/6 blocks, then simplify the sum to lowest terms.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: Subtract 1 1/2 - 3/4 by trading the whole and half into fourths (6/4 - 3/4). What fraction remains?

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: Compare 3/8 (31.5 mm) to 1/2 (42 mm). Which is smaller, and what is 3/8 written as a decimal (3 ÷ 8)?

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Find the area of a rectangle measuring 2/3 unit by 3/4 unit by multiplying (2/3) × (3/4) and simplifying.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

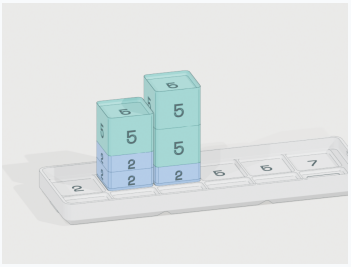
EXIT REFLECTION: Using the 84 mm Fraction Towers, explain why 1/2 + 1/4 equals 3/4 and NOT 2/6.

Postfix Evaluation, Factor Trees & Chained Operations

Hands-On Model: Prime Factor Studio (Logarithmic Prime Blocks) & 4-Register Stack • Standards: CCSS.MATH.5.OA.A.1, CCSS.MATH.6.EE.A.2

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Chained Operations: Evaluating (a + b) × (c - d) from the Inside Out):

Written algebra uses parentheses to show which inner operations must happen first. Evaluating from the inside out follows true mathematical structure: compute the first branch (6 + 1 = 7), hold that result while computing the second branch (9 - 5 = 4), and multiply the two branch values together.



Prime Factor Studio (p. viii) • Log Prime Towers

2. HANDS-ON BLOCKS & CONCEPT RULES: INSIDE-OUT EVALUATION & PRIME FACTOR TREE RULES

- Order of operations works from the inside out: evaluate inner groups, parentheses, and factor branches before combining them.
- Every composite number decomposes into a unique tower of prime factors (e.g., 28 = 2 × 2 × 7).
- In nested fractions such as (a − b) ÷ (c + d), the fraction bar groups the numerator and denominator as separate sub-problems.
- When evaluating two separate branches like (a + b) × (c − d), solve each branch first and then multiply their results.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Hands-On Setup: Use Prime Factor Studio blocks in the 12-socket comparator lid to model the two factor branches: (6 + 1) = [7] prime block and (9 - 5) = [2] + [2] = 4 tower, then combine their heights for 7 × 4 = 28. Question: Left sub-expression: 6 + 1 = 7. Right sub-expression: 9 - 5 = 4. Combined product: 7 × 4 = 28 (prime tower 2 × 2 × 7).

Verification Keystrokes: `6` `ENTER` `1` `+` `9` `ENTER` `5` `-` `×`

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
`6`	6	Value a (6) placed into Register X
`ENTER`	6	ENTER lifts 6 into Register Y
`1`	1	Value b (1) placed into Register X
`+`	7	Left operator (+): Combines Y(6) + X(1) = 7 in X
`9`	9	Value c (9) entered into X; 7 lifted to Y
`ENTER`	9	ENTER lifts 9 into Z and 7 into T
`5`	5	Value d (5) placed into Register X
`-`	4	Right operator (-): Combines Z/Y(9) - X(5) = 4 in X; 7 drops to Y
`×`	28	Root operator (×): Combines Y(7) × X(4) = 28 in X!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Evaluate (8 + 2) × (14 - 9) by finding the value of each parenthesized group first, then multiplying the two results.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): Evaluate (15 - 5) ÷ (3 + 2) by computing the numerator and denominator separately, then dividing.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 07 Lab: Postfix Evaluation, Factor Trees & Chained Operations

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your step-by-step paper math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

Investigation 1: Evaluate (4 + 3) × (10 - 6) by solving each parenthesized group first.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: Evaluate (20 - 4) ÷ (2 × 4) by finding the numerator and denominator first.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: Evaluate (18 - 6) ÷ (1 + 2) and write the prime factorization of the numerator.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Evaluate 5 + (3 × 4) by computing the inner product before adding 5.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

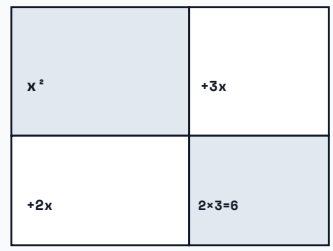
EXIT REFLECTION: In (8 + 2) × (9 − 4), where does the first branch result wait while you solve the second branch?

Algebraic Expressions, Base-x Polynomials & Quadratic Factoring

Hands-On Model: Prime Factor Studio (PRIME-01) & Algebraic Area Diagram • Standards: CCSS.MATH.7.EE.A.1, CCSS.MATH.8.EE.C.7

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ with Prime Factor Blocks):

A polynomial $ax^2 + bx + c$ mirrors base-10 place value with variable powers (x^2, x^1, x^0) in place of 10^2, 10^1, 10^0. Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ decomposes the constant term into prime factors ($2 \times 3 = 6$) whose sum equals the linear coefficient ($2 + 3 = 5$).



2. VISUAL MODEL & CONCEPT RULES: QUADRATIC AREA MODEL & PRIME FACTORING RULES

- A polynomial $ax^2 + bx + c$ mirrors base-10 place value with variable powers (x^2, x^1, x^0) in place of 10^2, 10^1, 10^0.
- To factor $x^2 + bx + c$ into $(x + p)(x + q)$, use Prime Factor blocks to factor c into two numbers p and q whose product is c and whose sum is b.
- When c is negative, p and q have opposite signs; when c is positive and b is negative, both p and q are negative.
- Zero-Product Property: The roots of $(x + p)(x + q) = 0$ are $x = -p$ and $x = -q$ (which you can check on the calculator with [POLY]).

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Setup: Stack Prime Factor blocks [2] and [3] ($2 \times 3 = 6$) in the comparator lid and sketch the 2×2 area grid for $(x + 2)(x + 3) = x^2 + 3x + 2x + 6$. Question: Prime factors of 6 are 2 and 3, and their sum is $2 + 3 = 5$. The factored form is $(x + 2)(x + 3)$ with roots x = -2 and x = -3.

Verification Keystrokes: 1 ENTER 5 ENTER 6 g POLY

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
1	1	Coefficient a = 1 entered into X
ENTER	1	1 lifted to Y
5	5	Coefficient b = 5 entered into X
ENTER	5	b in Y, a in Z
6	6	Coefficient c = 6 entered into X
g POLY	-2	Press [g:] [POLY]: Root 1 = -2 in Y (factor: x + 2), Root 2 = -3 in X (factor: x + 3)!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Factor $x^2 + 7x + 12$ by decomposing 12 with Prime Factor blocks to find the factor pair whose sum is 7, and find both roots of $x^2 + 7x + 12 = 0$.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Warm-Up B (Extension): Factor $x^2 - 5x + 6$ into two binomials by finding two negative factors of +6 whose sum is -5, and state both roots.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Unit 08 Lab: Algebraic Expressions, Base-x Polynomials & Quadratic Factoring

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the geometric/coordinate model in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Factor $x^2 + 6x + 8$ by decomposing 8 into a factor pair that sums to 6, and find the roots.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Factor $x^2 - 4$ (difference of two squares, where the linear coefficient $b = 0$).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Factor $x^2 + 5x + 4$ into two binomials and state the solutions to $x^2 + 5x + 4 = 0$.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Factor $x^2 - 5x + 6$ using signed factor pairs and state both roots.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

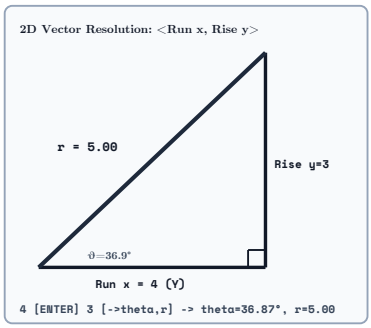
EXIT REFLECTION: For $x^2 + bx + c = (x + p)(x + q)$, how are the numbers p and q related to b and c?

Trigonometry, Vectors & Polar/Rectangular Coordinates

Hands-On Model: Right-Triangle & 2D Vector Diagram (Optional Ratio Check: Fraction Studio) • Standards: CCSS.MATH.HSG.SRT.C.8, CCSS.MATH.HSN.VM.A.1

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Vector Magnitude & Direction: Rectangular to Polar Coordinates):

Every 2D vector or right triangle relates two perpendicular legs (dx, dy) to a hypotenuse r and angle ϑ via $r = \sqrt{(x^2 + y^2)}$ and $\tan(\vartheta) = y/x$. Keeping horizontal component dx in Register Y and vertical component dy in Register X pairs the two coordinates naturally.



2. VISUAL MODEL & CONCEPT RULES: RIGHT-TRIANGLE RATIOS & 2D VECTOR DECOMPOSITION RULES

- Trigonometric functions are ratios of right-triangle sides: sin(ϑ) = Opposite / Hypotenuse, cos(ϑ) = Adjacent / Hypotenuse, tan(ϑ) = Opposite / Adjacent.
- Every 2D vector v = <dx, dy> forms a right triangle with magnitude $r = \sqrt{(dx^2 + dy^2)}$ and direction angle ϑ = arctan(dy / dx).
- To add two vectors A = <x1, y1> and B = <x2, y2>, add their horizontal components (x1 + x2) and vertical components (y1 + y2) separately.
- Stack Check: Place horizontal component dx in Y and vertical component dy in X, then press [f:] [→ϑ,r] to verify ϑ and r!

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Setup: Draw a right triangle with horizontal run dx = 4 (Register Y) and vertical rise dy = 3 (Register X) to model the 3-4-5 vector. Question: Run = 4, Rise = 3. Predict magnitude: $r = \text{sqrt}(4^2 + 3^2) = 5$. Predict angle: arctan(3/4) ~ 36.87°.

Verification Keystrokes: [4] [ENTER] [3] [↶] [▸θ,r] [x⇄y]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[4]	4	Enter horizontal component x = 4
[ENTER]	4	4 lifted to register Y
[3]	3	Enter vertical component y = 3 in X
[↶] [▸θ,r]	36.87	Rectangular to Polar! Screen displays angle ϑ = 36.87°
[x⇄y]	5.00	Swap registers: Y displays vector magnitude r = 5.00!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): A surveyor walks 60 m East (dx = 60) and 80 m North (dy = 80). Use the Pythagorean theorem and tangent ratio to find the straight-line distance r and angle ϑ north of east.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Warm-Up B (Extension): Two cables pull on a post: Vector A is <5, 2> and Vector B is <-2, 4>. Find the resultant vector <x_net, y_net> and its magnitude r.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Unit 09 Lab: Trigonometry, Vectors & Polar/Rectangular Coordinates

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the geometric/coordinate model in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Find the magnitude r and direction angle ϑ for the vector <12, 5>.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Find the hypotenuse of a right triangle with legs of length 7 and 24.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: In a 30°-60°-90° triangle, the side opposite 30° is half the hypotenuse (42 mm vs. 84 mm). What is sin(30°)?

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: A vector has magnitude r = 10 at angle ϑ = 45°. Find its horizontal (x = r cos ϑ) and vertical (y = r sin ϑ) components.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How are the horizontal run (dx) and vertical rise (dy) of a vector related to its magnitude r?

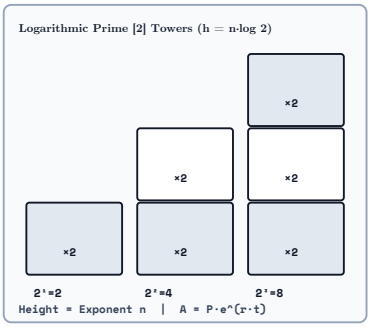
UNIT 10 GUIDED CONCEPT LAB • GRADES 10–12 (SC-32)

Exponential Growth, Compound Interest & Logarithms

Hands-On Model: Prime Factor Studio (PRIME-01 Logarithmic Blocks) & Growth Curve • Standards: CCSS.MATH.HSF.LE.A.1, CCSS.MATH.HSF.LE.A.4

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Continuous Compounding & Euler's Constant e):

Exponential growth P·e^(rt) compounds continuously by scaling a base quantity according to an inner rate-time product (r × t). Evaluating from the inside out—computing the exponent r × t first, raising the base, and multiplying by P—follows the exact mathematical hierarchy without parentheses.



2. VISUAL MODEL & CONCEPT RULES: EXPONENTIALS, CONTINUOUS COMPOUNDING & DOUBLING TIME

- Discrete exponential growth multiplies by a constant factor each period: A = P × b^t (or A = P × (1 + r/n)^(n·t)).
- Continuous exponential growth follows A = P × e^(r·t), where Euler's constant e ~ 2.71828 is the base of natural growth.
- Logarithms invert exponentials (ln(e^x) = x) and turn multiplication into addition: ln(a × b) = ln(a) + ln(b).
- To find continuous doubling time, solve e^(r·t) = 2 by taking the natural logarithm: t = ln(2) / r ~ 0.6931 / r.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Setup: Stack Prime Factor [2] blocks (15 mm each) to show linear height addition representing exponential doubling ($2^1 = 2$, $2^2 = 4$, $2^3 = 8$). Question: Invest $1,000 at 7% continuous interest for 10 years: A = 1000 × e^(0.07 × 10) = 1000 × e^0.7. Will it double? (e^0.7 ~ 2.01).

Verification Keystrokes: `0` `.` `0` `7` `ENTER` `1` `0` `×` `e^x` `1` `0` `0` `0` `×`

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
`0` `.` `0` `7`	0.07	Rate r = 0.07 entered in X
`ENTER`	0.07	Rate lifted to Y
`1` `0`	10	Time t = 10 entered in X
`×`	0.7	Exponent r × t = 0.70
`e^x`	2.0138	e^0.70 = 2.01375 (Growth factor: more than doubled!)
`1` `0` `0` `0`	1000	Principal P = 1000 (growth factor lifts to Y)
`×`	2013.75	Total accumulated amount A = $2,013.75!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): A bacteria culture starts with P = 500 cells and triples every hour (A = 500 × 3^t). Calculate the population after t = 4 hours.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Warm-Up B (Extension): A radioactive isotope decays according to N(t) = N_0 · e^(-0.035t). Find its half-life t = ln(2) / 0.035 to two decimal places.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Unit 10 Lab: Exponential Growth, Compound Interest & Logarithms

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the geometric/coordinate model in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Evaluate e^2 to 4 decimal places.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Find the continuous doubling time t = ln(2) / r for an investment growing at r = 8% (0.08).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Evaluate 10^3.5 (note that 10^3 < 10^3.5 < 10^4).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Solve 10^x = 500 for x using the base-10 logarithm log10(500).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

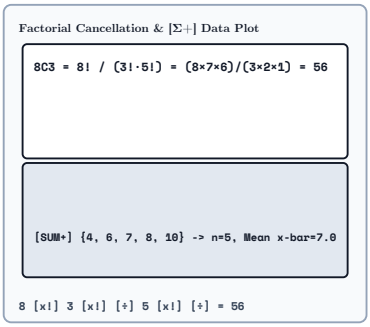
EXIT REFLECTION: Why is the natural logarithm [LN] the inverse operation needed to solve for doubling time t?

Combinatorics, Probability Ratios & Series Accumulation

Hands-On Model: Prime Factor Studio (PRIME-01 Factorial Cancellation) & Statistical Data Plot • Standards: CCSS.MATH.HSS.CP.B.9, CCSS.MATH.HSA.SSE.B.4

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Combinations vs. Permutations: Subsets & Prime Cancellation):

Combinations nCr = n! / (r!(n-r)!) count unordered selections by dividing ordered permutations by the r! duplicate arrangements of the chosen group. Canceling 5! from 8! leaves (8 × 7 × 6) / 3! = 56.



2. VISUAL MODEL & CONCEPT RULES: COMBINATIONS, PERMUTATIONS & SAMPLE STATISTICS RULES

- Factorial n! = n × (n − 1) × ... × 1 counts the ways to arrange n distinct items; Prime Factor blocks show how common factors cancel in n! / r!.
- Permutations (order matters): nPr = n! / (n − r)! counts ordered arrangements of r items chosen from n.
- Combinations (order does not matter): nCr = n! / (r! × (n − r)!) divides out the r! duplicate orderings.
- Sample Mean: x-bar = (sum of values) / n. You can check factorials with [f:][x!] and sample statistics with [Σ+].

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Setup: Use Prime Factor blocks to model prime factor cancellation in 8! / (3! × 5!) = (8 × 7 × 6) / (3 × 2 × 1) = 8 × 7 = 56. Question: How many ways to choose a committee of 3 from 8 candidates? 8C3 = 8! / (3! × 5!) = (8 × 7 × 6) / 6 = 56.

Verification Keystrokes: 8 ↶ x! 3 ↶ x! ÷ 5 ↶ x! ÷

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
8	8	Total candidates n = 8
↶ x!	40320	8! = 40,320 in register X
3	3	Subset r = 3 (40320 lifts to Y)
↶ x!	6	3! = 6 in X
÷	6720	40320 / 6 = 6720 in X
5	5	Remaining n - r = 5 (6720 lifts to Y)
↶ x!	120	5! = 120 in X
÷	56	6720 / 120 = 56 combinations!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): A lottery draws 4 winning numbers from a pool of 10 (order does not matter). Simplify 10C4 = 10! / (4! × 6!) by canceling 6! first, then check your result.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Warm-Up B (Extension): Given the four data points {12, 15, 18, 21}, find the sample size n, the total sum of the values, and the sample mean x-bar.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Unit 11 Lab: Combinatorics, Probability Ratios & Series Accumulation

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the geometric/coordinate model in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Evaluate 6! (the number of ways to arrange 6 distinct books on a shelf).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: How many ways can 5 runners finish in 1st, 2nd, and 3rd place? (Compute 5P3 = 5! / 2!).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: How many 2-person partnerships can be formed from 7 students? (Compute 7C2 = 7! / (2! × 5!)).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Find the mean x-bar of the two data values 5 and 15.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

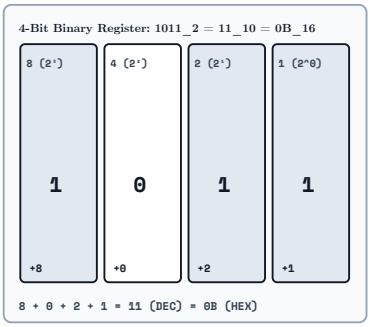
EXIT REFLECTION: Why does a Combination C(n, r) produce fewer outcomes than a Permutation P(n, r)?

Digital Logic, Base Conversions & Computer Architecture

Hands-On Model: Place Value Studio (Binary [0]/[1] Register Trays & Base-N Columns) • Standards: CCSS.MATH.PRACTICE.MP5, CCSS.MATH.HSA.SSE.A.1

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Positional Number Systems: Decimal, Binary & Hexadecimal):

Binary (base-2) and hexadecimal (base-16) follow the exact same positional place-value principle as base-10, using powers of 2 (8, 4, 2, 1) or 16 instead of powers of 10. Every 4-bit binary group (0000 to 1111) maps to exactly one hexadecimal digit (0 to F).



2. VISUAL MODEL & CONCEPT RULES: POSITIONAL NUMBER BASES & BITWISE LOGIC RULES

• Binary (Base 2) uses powers of 2 (..., 8, 4, 2, 1) with digits 0 and 1; Hexadecimal (Base 16) groups 4 binary bits into a single digit 0–9 or A–F.

• Bitwise AND (&): output bit is 1 only when BOTH input bits in that column are 1 (used for masking and network subnets).

• Bitwise OR (|): output bit is 1 when EITHER input bit is 1. Bitwise XOR (^): output bit is 1 when the two input bits DIFFER.

• Stack Check: Press [f:] [BASE] to verify conversions between DEC, HEX, BIN, and OCT, or use [AND], [OR], [XOR] on two integers.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Setup: Use the sockets of the Place Value Tray as binary bit positions (8, 4, 2, 1). Place [1] Digit Blocks for active '1' bits and [0] Digit Blocks for '0' bits. Question: Place [1], [0], [1], [1] in the 8, 4, 2, 1 sockets: binary 1011_2 = 8 + 0 + 2 + 1 = 11_10. In hexadecimal (A=10, B=11), 11_10 = 0B_16.

Verification Keystrokes: [1] [1] [↖] [BASE]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[1] [1]	11	Enter decimal integer 11 in DEC mode
[↖] [BASE]	HEX 0B	Press [f:] [BASE]: screen displays Hexadecimal 0B!
[↖] [BASE]	BIN 1011	Press [f:] [BASE]: screen displays Binary 1011!
[↖] [BASE]	OCT 13	Press [f:] [BASE]: screen displays Octal 13!
[↖] [BASE]	DEC 11	Press [f:] [BASE]: returns to Decimal 11

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Express decimal 255 as a sum of powers of 2 (128 + 64 + 32 + 16 + 8 + 4 + 2 + 1) to write its 8-bit binary and 2-digit hexadecimal representations.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Warm-Up B (Extension): Compare 12 (1100_2) and 3 (0011_2) column by column to find the result of 12 OR 3 in both binary and decimal.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Unit 12 Lab: Digital Logic, Base Conversions & Computer Architecture

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the geometric/coordinate model in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Decompose decimal 42 into powers of 2 (32, 16, 8, 4, 2, 1) to write its 6-bit binary representation.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Convert hexadecimal 2A_16 to decimal using place value ($2 \times 16^1 + 10 \times 16^0$).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Evaluate the bitwise AND of 15 (1111_2) and 7 (0111_2) in binary and decimal.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Evaluate the bitwise XOR of 12 (1100_2) and 10 (1010_2) in binary and decimal.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: Why does a single hexadecimal digit (0–F) correspond to exactly 4 binary bits (one nybble)?

Student Self-Check Answer Key

Check your Warm-Up and Investigation answers after completing each unit. Full step-by-step traces are in Book 1.

Unit 01: Cardinality, Counting & Step Accumulation (Pre-K – Kindergarten)
Warm-Up A: 5 ([5] Digit Block; 4 + 1 = 5) | B: Row 1 has 4 more (6 - 2 = 4)

Investigation 1: 5 • **Investigation 2: 3**
Investigation 3: 0 (Zero) • **Investigation 4: 4**

Unit 02: Part-Part-Whole Number Bonds & Missing Addends (Grades K–1)
Warm-Up A: 9 ([9] Digit Block; 5 + 4 = 9) | B: Missing Part is 3 (8 - 5 = 3)

Investigation 1: 4 • **Investigation 2: 9**
Investigation 3: 9 • **Investigation 4: 3**

Unit 03: Complements to 10 & Gold [+1] Carry Regrouping (Grade 1)
Warm-Up A: 4 ones (10 - 6 = 4) | B: 15 (9 + 1 = 10, then 10 + 5 = 15)

Investigation 1: 6 • **Investigation 2: 12**
Investigation 3: 1 • **Investigation 4: 14**

Unit 04: Twin-Row Place Value Tray: Doubles & Equal Groups (Grades 1–2)
Warm-Up A: 14 ([1] Ten and [4] Ones on the tray; 7 + 7 = 14) | B: 13 (6 + 6 = 12, then 12 + 1 = 13)

Investigation 1: 16 • **Investigation 2: 17**
Investigation 3: 18 • **Investigation 4: 11**

Unit 05: Place Value & The 4-Level Stack Elevator (Grades 3–4)
Warm-Up A: 3 cookies per bag ((15 - 6) ÷ 3 = 3) | B: 19 (24 ÷ 6 = 4, then 4 + 15 = 19)

Investigation 1: 27 • **Investigation 2: 6**
Investigation 3: 4 • **Investigation 4: 3 eggs**

Unit 06: Fraction Studio: 84 mm Towers & Two-Dot Entry (Grades 4–5)
Warm-Up A: 3/4 (three 21 mm fourth-blocks = 63 mm) | B: 1/4 (6/24 = 1/4, height 21 mm)

Investigation 1: 1/2 (42 mm) • **Investigation 2: 3/4 (63 mm)**
Investigation 3: 0.375 • **Investigation 4: 1/2 (42 mm)**

Unit 07: Postfix Trees, Order of Operations & Prime Factors (Grades 5–6)
Warm-Up A: (8 + 2) = 10 and (14 - 9) = 5; 10 × 5 = 50 (Prime factors: 2 × 5 × 5) | B: Numerator = 10, Denominator = 5; 10 ÷ 5 = 2

Investigation 1: 7 × 4 = 28 • **Investigation 2: 16 ÷ 8 = 2**
Investigation 3: 12 ÷ 3 = 4 (12 = 2×2×3) • **Investigation 4: 5 + 12 = 17**

Unit 08: Base-x Polynomials, Prime Factor Pairs & Roots (Grades 7–8)
Warm-Up A: 3 × 4 = 12 and 3 + 4 = 7; factored form (x + 3)(x + 4), roots x = -3 and x = -4 | B: (-2) × (-3) = +6 and (-2) + (-3) = -5; factored form (x - 2)(x - 3), roots

Investigation 1: (x + 2)(x + 4); roots -2, -4 • **Investigation 2: (x - 2)(x + 2); roots 2, -2**
Investigation 3: (x + 1)(x + 4); roots -1, -4 • **Investigation 4: (x - 2)(x - 3); roots 2, 3**

Unit 09: Coordinate Geometry, Trigonometry & Vectors (Grades 9–11)
Warm-Up A: r = √(60² + 80²) = 100 m; ϑ = arctan(80/60) ˜ 53 | B: Resultant <3, 6>; Magnitude r = √(3² + 6²) = √45 ˜ 6

Investigation 1: r = 13, ϑ = 22.62° • **Investigation 2: r = √(49 + 576) = 25**
Investigation 3: 42 / 84 = 0.5 • **Investigation 4: x ˜ 7.07, y ˜ 7.07**

Unit 10: Exponents, Logarithms & Continuous Growth (Grades 10–12)
Warm-Up A: 3^4 = 81; 500 × 81 = 40,500 bacteria | B: t = ln(2) / 0

Investigation 1: 7.3891 • **Investigation 2: 8.66 years**
Investigation 3: 3162.28 • **Investigation 4: 2.6990**

Unit 11: Combinatorics, Probability Ratios & Series Statistics (Grades 11–12 / AP)
Warm-Up A: (10 × 9 × 8 × 7) / (4 × 3 × 2 × 1) = 5040 / 24 = 210 combinations | B: Count n = 4, Sum = 66, Mean x-bar = 66 / 4 = 16

Investigation 1: 6 × 5 × 4 × 3 × 2 × 1 = 720 • **Investigation 2: 5 × 4 × 3 = 60**
Investigation 3: (7 × 6) / 2 = 21 • **Investigation 4: (5 + 15) / 2 = 10**

Unit 12: Digital Logic, Base-N & Computer Architecture (Grades 9–12 / AP CS)
Warm-Up A: Binary: 11111111_2 (eight 1s); Hexadecimal: FF_16 (15 × 16 + 15 = 255) | B: 1100_2 OR 0011_2 = 1111_2 = 15_10

Investigation 1: 32 + 8 + 2 → 101010_2 • **Investigation 2: 32 + 10 = 42**
Investigation 3: 0111_2 = 7 • **Investigation 4: 0110_2 = 6**

Certificate of Hands-On Stack Mastery

Awarded for completing Units 01–12 of the StackCalc Hands-On Math Lab Workbook.

CERTIFICATE OF HANDS-ON MATHEMATICS MASTERY

This certifies that: _____

has completed the 12 Hands-On Math Lab investigations from Counting & Number Bonds through Fractions, Algebraic Factoring, Trigonometry, Logarithms, Probability, and Digital Logic—building physical block models first and verifying every solution on the 4-level StackCalc RPN stack (T, Z, Y, X).

[ ] Elementary Stack Explorer (Units 01–07 • SC-6) **[ ] Scientific Stack Master (Units 08–12 • SC-32)**

Teacher / Parent Signature: _____ Date: _____

ESSENTIAL SC-6 & SC-32 KEYCAP SEQUENCES TO REMEMBER

Doubling Any Number in Register X (e.g., 2 × 7 = 14)	
Parenthesis-Free Chained Math: (8 + 2) × (9 − 4) = 50	
Two-Dot Fraction Addition: 3/4 + 1/2 = 1 1/4	
Quadratic Roots of $x^2 - 5x + 6 = 0$ (Roots: 2, 3)	
Pythagorean Hypotenuse of 3 and 4 → 5	